COMPUTER ENGINEERING

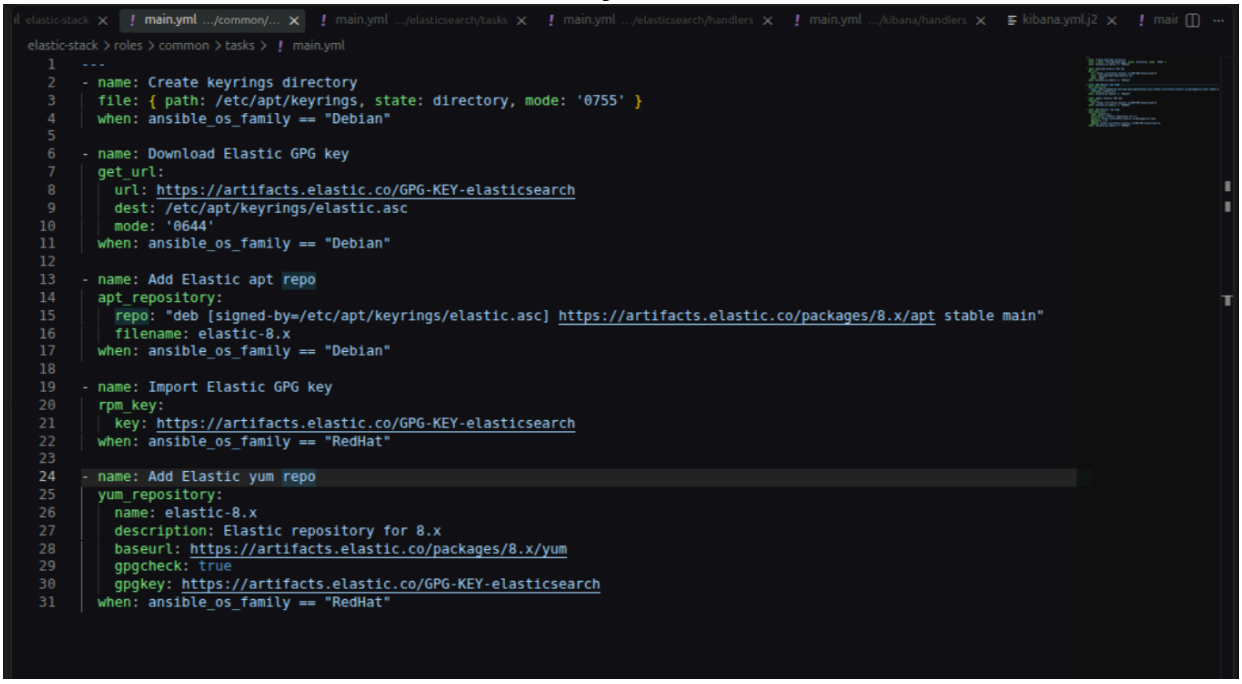
QUEZON CITY

COURSE CODE/TITLE:	CPE212	**SECTION:**	CPE31S2
DATE PERFORMED:	10/05/26	**DATE SUBMITTED:**	10/05/26
NAME:	Ivan Wayne San Juan	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY 10.1 **Install, Configure and Manage Enterprise Log Monitoring via Ansible**

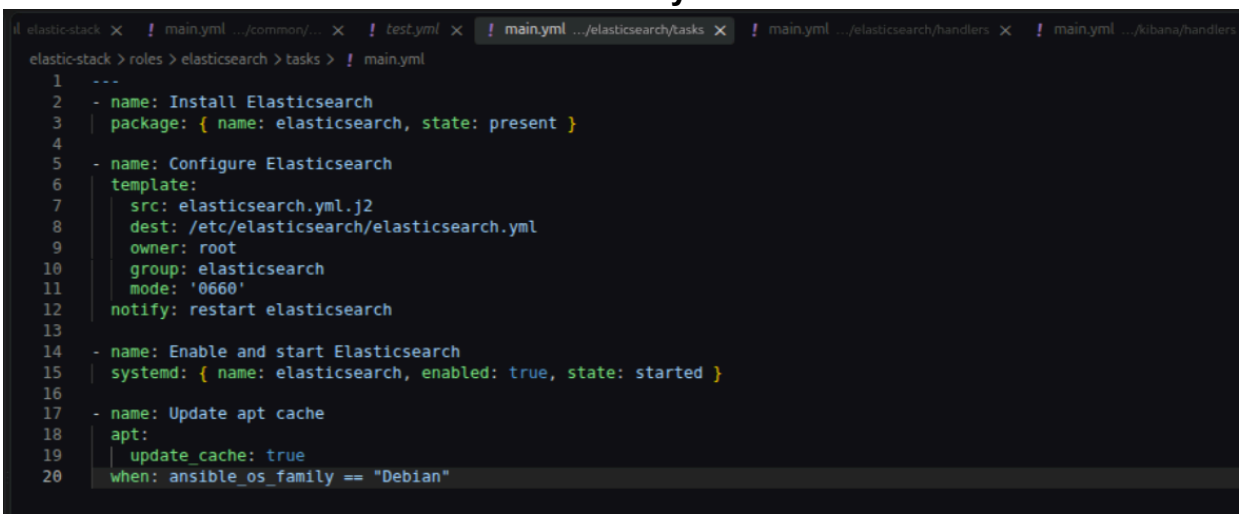
1. PROCEDURE OUTPUT

1. Create a playbook that:
a. Install and configure Elastic Stack in separate hosts (Elastic Search, Kibana, Logstash)
Code:
COMMON:
elastic-stack/roles/common/tasks/main.yml:



ELASTIC SEARCH

elastic-stack/roles/elasticsearch/tasks/main.yml:



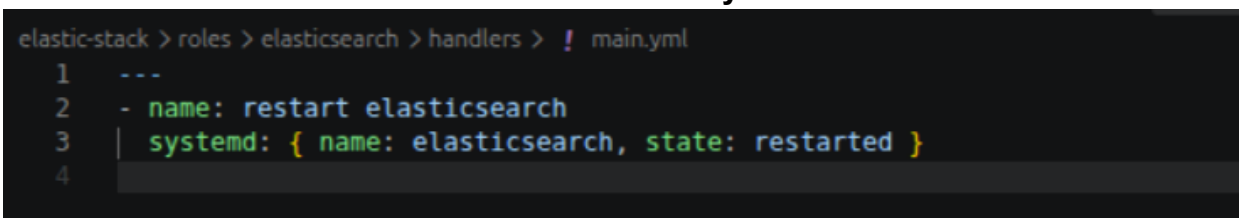
```yaml
---
- name: Install Elasticsearch
  package: { name: elasticsearch, state: present }

- name: Configure Elasticsearch
  template:
    src: elasticsearch.yml.j2
    dest: /etc/elasticsearch/elasticsearch.yml
    owner: root
    group: elasticsearch
    mode: '0660'
  notify: restart elasticsearch

- name: Enable and start Elasticsearch
  systemd: { name: elasticsearch, enabled: true, state: started }

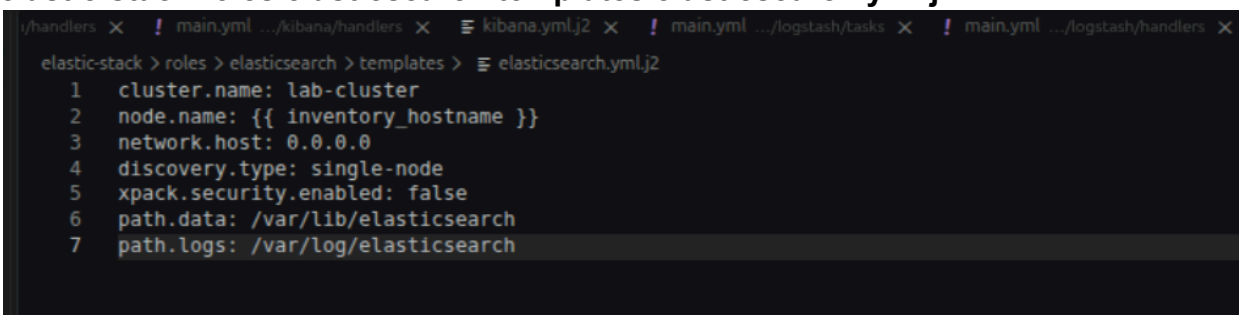
- name: Update apt cache
  apt:
    update_cache: true
  when: ansible_os_family == "Debian"
```

elastic-stack/roles/elasticsearch/handlers/main.yml:



```yaml
---
- name: restart elasticsearch
  systemd: { name: elasticsearch, state: restarted }
```

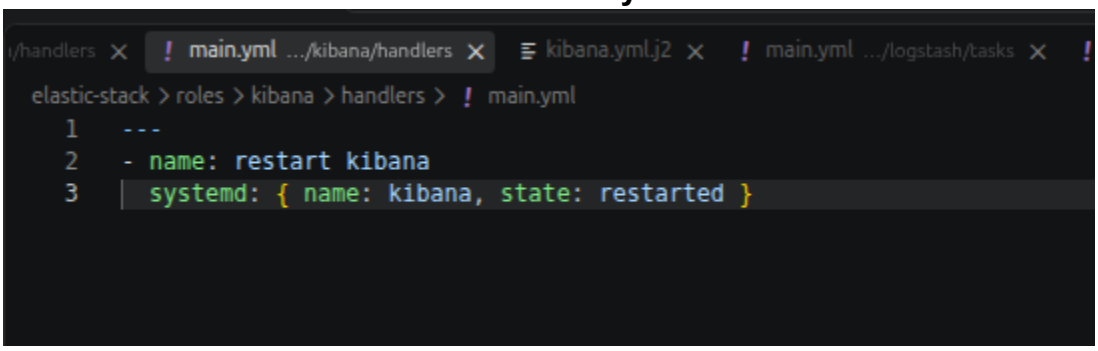
elastic-stack/roles/elasticsearch/templates/elasticsearch.yml.j2:



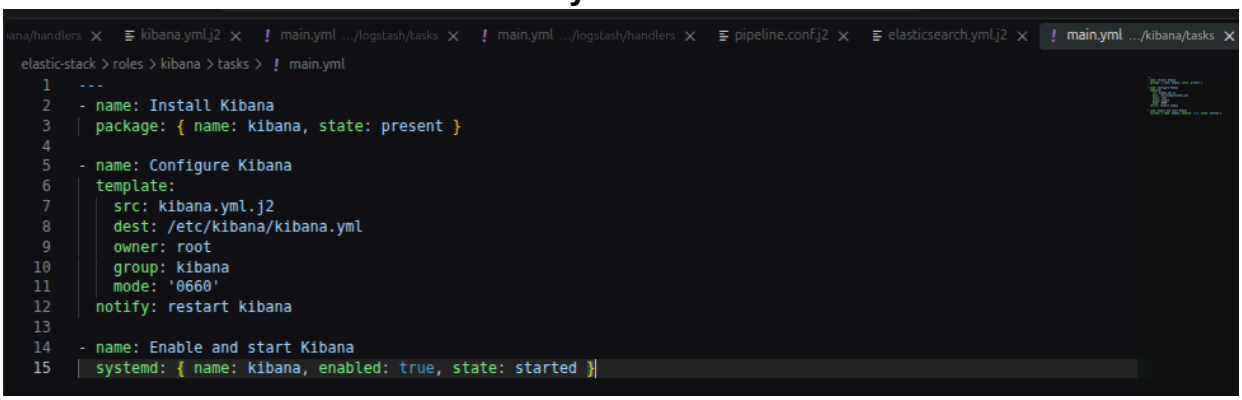
```
cluster.name: lab-cluster
node.name: {{ inventory_hostname }}
network.host: 0.0.0.0
discovery.type: single-node
xpack.security.enabled: false
path.data: /var/lib/elasticsearch
path.logs: /var/log/elasticsearch
```

KIBANA

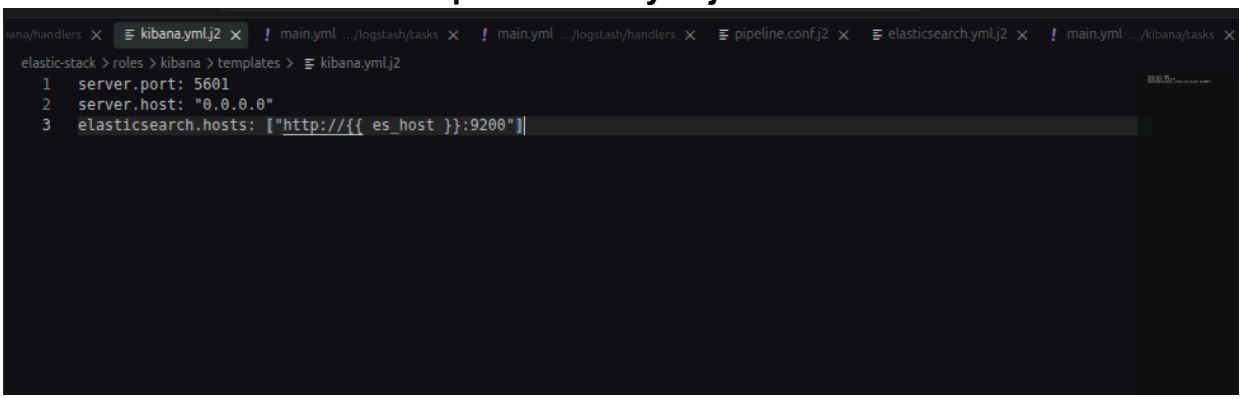
elastic-stack/roles/kibana/handlers/main.yml:



elastic-stack/roles/kibana/tasks/main.yml

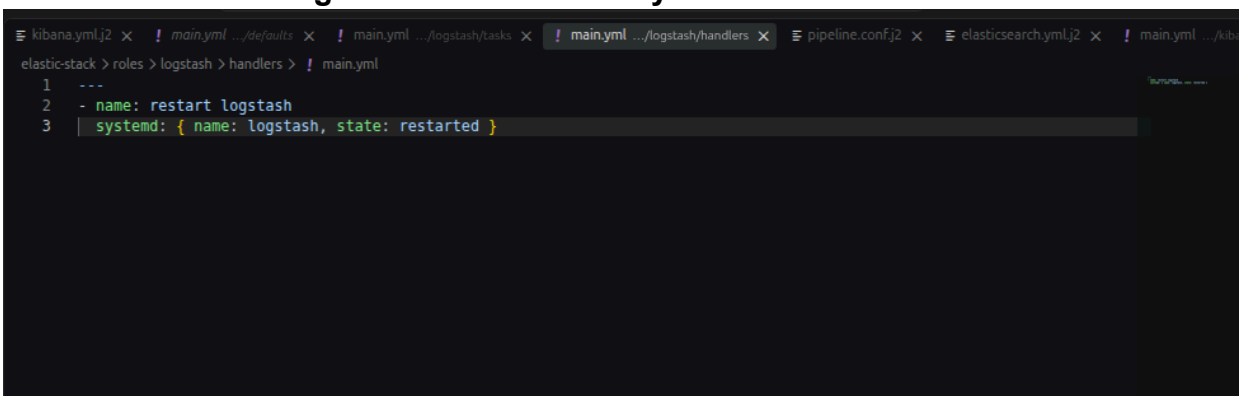


elastic-stack/roles/kibana/template/kibana.yml.j2:

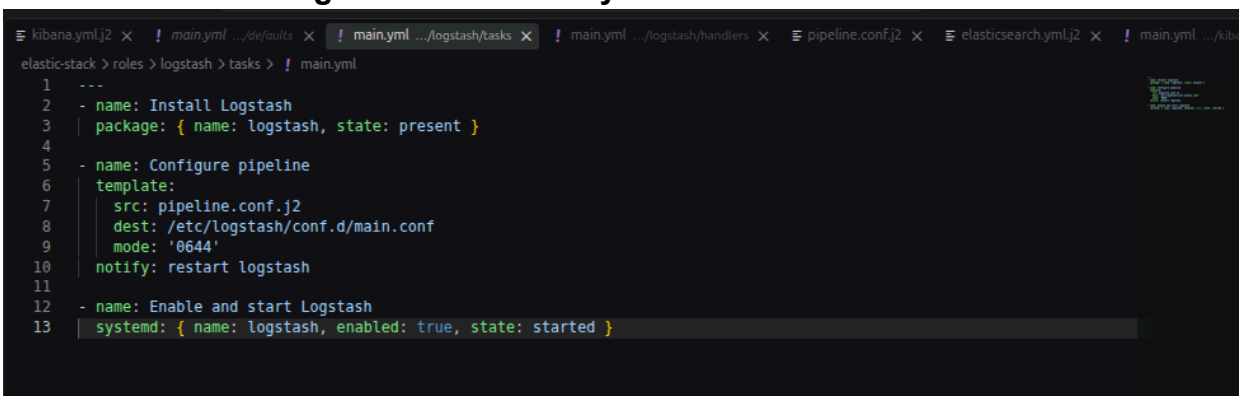


LOGSTASH

elastic-stack/roles/logstash/handlers/main.yml:



elastic-stack/roles/logstash/tasks/main.yml:



elastic-stack/roles/logstash/template/pipeline.yml.j2



Ubuntu:

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab/elastic-stack$ ansible-playbook -i inventories/ubuntu.ini site.yml -K -b
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.103]
ok: [192.168.56.102]

TASK [common : Create keyrings directory] **************************************
ok: [192.168.56.103]
ok: [192.168.56.102]

TASK [common : Download Elastic GPG key] ***************************************
ok: [192.168.56.103]
ok: [192.168.56.102]

TASK [common : Add Elastic apt repo] *******************************************
ok: [192.168.56.103]
ok: [192.168.56.102]

TASK [common : Import Elastic GPG key] *****************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]

TASK [common : Add Elastic yum repo] *******************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]

PLAY [elasticsearch] ***********************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.102]

TASK [elasticsearch : Install Elasticsearch] ***********************************
ok: [192.168.56.102]

TASK [elasticsearch : Configure Elasticsearch] *********************************
ok: [192.168.56.102]

TASK [elasticsearch : Enable and start Elasticsearch] **************************
ok: [192.168.56.102]

TASK [elasticsearch : Update apt cache] ****************************************
changed: [192.168.56.102]

PLAY [kibana] ******************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.103]
```

```
TASK [Gathering Facts] *********************************************************
ok: [192.168.56.102]

TASK [elasticsearch : Install Elasticsearch] ***********************************
ok: [192.168.56.102]

TASK [elasticsearch : Configure Elasticsearch] *********************************
ok: [192.168.56.102]

TASK [elasticsearch : Enable and start Elasticsearch] **************************
ok: [192.168.56.102]

TASK [elasticsearch : Update apt cache] ****************************************
changed: [192.168.56.102]

PLAY [kibana] ******************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.103]

TASK [kibana : Install Kibana] *************************************************
ok: [192.168.56.103]

TASK [kibana : Configure Kibana] ***********************************************
ok: [192.168.56.103]

TASK [kibana : Enable and start Kibana] ****************************************
ok: [192.168.56.103]

PLAY [logstash] ****************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.103]

TASK [logstash : Install Logstash] *********************************************
changed: [192.168.56.103]

TASK [logstash : Configure pipeline] *******************************************
changed: [192.168.56.103]

TASK [logstash : Enable and start Logstash] ************************************
changed: [192.168.56.103]

RUNNING HANDLER [logstash : restart logstash] **********************************
changed: [192.168.56.103]

PLAY RECAP *********************************************************************
192.168.56.102             : ok=9    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.103             : ok=13   changed=4    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab/elastic-stack$ ansible -i inventories/ubuntu.ini elasticsearch -m command -a "dpkg -s elasticsearch" -K
BECOME password:
192.168.56.102 | CHANGED | rc=0 >>
Package: elasticsearch
Status: install ok installed
Priority: optional
Section: web
Installed-Size: 1279587
Maintainer: Elasticsearch Team <info@elastic.co>
Architecture: amd64
Source: elasticsearch
Version: 8.19.22
Depends: bash (>= 4.1), libc6, adduser, coreutils (>= 8.4)
Conflicts: elasticsearch-oss
Conffiles:
 /etc/elasticsearch/elasticsearch.yml 5de21e0eabb17661972c3474282d9c9d
 /etc/elasticsearch/elasticsearch-plugins.example.yml 7f8a7b373bc214a1ed991c0599481abe
 /etc/elasticsearch/jvm.options 34096a8edc090629569095cd306bf8ff
 /etc/elasticsearch/log4j2.properties 797d6b5656abef1b32bfd37d494870df
 /etc/elasticsearch/role_mapping.yml 611f6a6b8045d3c6ef65e4e4516e5045
 /etc/elasticsearch/roles.yml 7abe692d5407592c6eede1767c5f3e18
 /etc/elasticsearch/users d41d8cd98f00b204e9800998ecf8427e
 /etc/elasticsearch/users_roles d41d8cd98f00b204e9800998ecf8427e
 /etc/default/elasticsearch e8663c0646f3409dec22579c24d0a646
Description: Distributed RESTful search engine built for the cloud
 Reference documentation can be found at
   https://www.elastic.co/guide/en/elasticsearch/reference/current/index.html
   and the 'Elasticsearch: The Definitive Guide' book can be found at
   https://www.elastic.co/guide/en/elasticsearch/guide/current/index.html
Homepage: https://www.elastic.co/
License: Elastic-License
```

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab/elastic-stack$ ansible -i inventories/ubuntu.ini kibana -m command -a "dpkg -s kibana" -K
BECOME password:
192.168.56.103 | CHANGED | rc=0 >>
Package: kibana
Status: install ok installed
Priority: optional
Section: default
Installed-Size: 1685060
Maintainer: Kibana Team <info@elastic.co>
Architecture: amd64
Version: 8.19.22
Depends: adduser
Conffiles:
 /etc/default/kibana a865723ccdd218836f6afbb4c619c447
 /etc/kibana/kibana.yml 436ca5f905c9038304f525641f2fc5e5
 /etc/kibana/node.options a34b8fe9075b2c6f20578760bf354127
Description: Explore and visualize your Elasticsearch data
License: Elastic-License
Vendor: Elasticsearch, Inc.
Homepage: https://www.elastic.co
```

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab/elastic-stack$ ansible -i inventories/ubuntu.ini logstash -m command -a "dpkg -s logstash" -K
BECOME password:
192.168.56.103 | CHANGED | rc=0 >>
Package: logstash
Status: install ok installed
Priority: optional
Section: default
Installed-Size: 717516
Maintainer: <buildkite-agent@bk-agent-prod-gcp-1789751424582825685>
Architecture: amd64
Version: 1:8.19.22-1
Replaces: logstash (<< 7.10.0)
Conflicts: logstash (<< 7.10.0)
Conffiles:
 /etc/default/logstash a52e0fcdeca70519e18c376079d22478
 /etc/logstash/jvm.options 19d24698e660ea052c58e0d29c824770
 /etc/logstash/log4j2.properties 40212b07d14616e0876c02a99eb6f03b
 /etc/logstash/logstash-sample.conf 4ef421e7ee74e023c9b07f200cc6010a
 /etc/logstash/logstash.yml 581a0d2f2c3456ff6db38d589d7a353f
 /etc/logstash/pipelines.yml f4fec5524ff51ccd6a5e7f686cb8c3f5
 /etc/logstash/startup.options 63ffc442d4f1c1a56c178a1d8af85eb0
 /lib/systemd/system/logstash.service 21cc1a14599cae323b5751bfc2ce927e
Description: An extensible logging pipeline
License: Elastic-License
Vendor: Elastic
Homepage: https://www.elastic.co/logstash
```

Centos:

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab/elastic-stack$ ansible-playbook -i inventories/centos.ini site.yml -K -b
BECOME password:

PLAY [all] *********************************************************************************************************************************************************
*********************************************

TASK [Gathering Facts] *********************************************************************************************************************************************
*********************************************
ok: [192.168.56.106]

TASK [common : Create keyrings directory] **************************************************************************************************************************
*********************************************
skipping: [192.168.56.106]

TASK [common : Download Elastic GPG key] ***************************************************************************************************************************
*********************************************
skipping: [192.168.56.106]

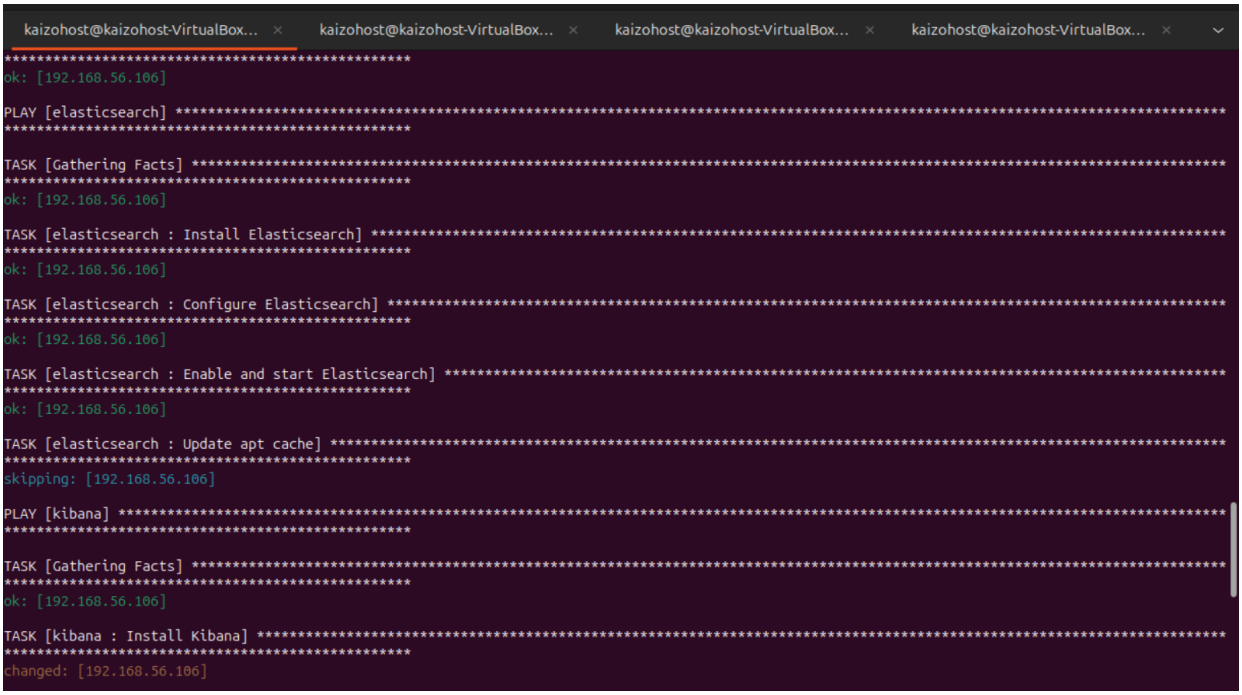
TASK [common : Add Elastic apt repo] *******************************************************************************************************************************
*********************************************
skipping: [192.168.56.106]

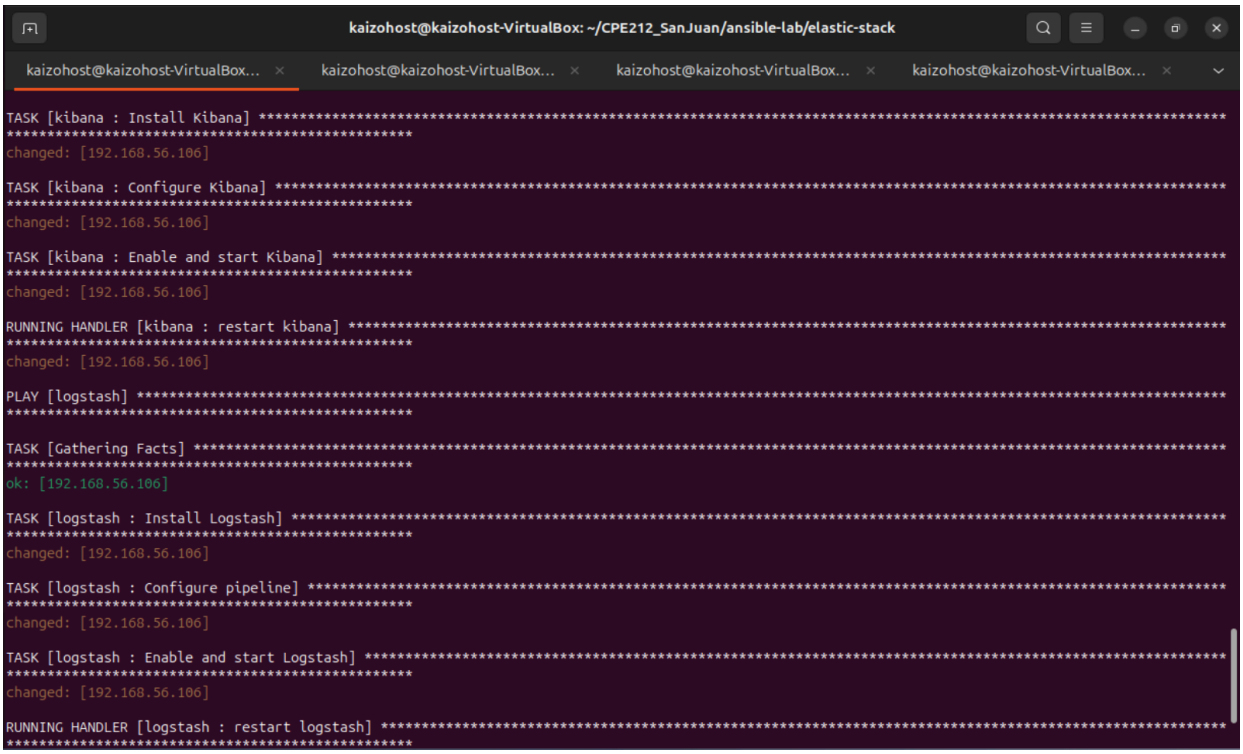
TASK [common : Import Elastic GPG key] *****************************************************************************************************************************
*********************************************
ok: [192.168.56.106]

TASK [common : Add Elastic yum repo] *******************************************************************************************************************************
*********************************************
ok: [192.168.56.106]

PLAY [elasticsearch] ***********************************************************************************************************************************************
*********************************************

TASK [Gathering Facts] *********************************************************************************************************************************************
*********************************************
ok: [192.168.56.106]
```


kaizohost@kaizohost-VirtualBox...
kaizohost@kaizohost-VirtualBox...
kaizohost@kaizohost-VirtualBox...
kaizohost@kaizohost-VirtualBox...
****
ok: [192.168.56.106]
PLAY [elasticsearch] **********
****
TASK [Gathering Facts] **********
****
ok: [192.168.56.106]
TASK [elasticsearch : Install Elasticsearch] **********
****
ok: [192.168.56.106]
TASK [elasticsearch : Configure Elasticsearch] **********
****
ok: [192.168.56.106]
TASK [elasticsearch : Enable and start Elasticsearch] **********
****
ok: [192.168.56.106]
TASK [elasticsearch : Update apt cache] **********
****
skipping: [192.168.56.106]
PLAY [kibana] **********
****
TASK [Gathering Facts] **********
****
ok: [192.168.56.106]
TASK [kibana : Install Kibana] **********
****
changed: [192.168.56.106]


kaizohost@kaizohost-VirtualBox: ~/CPE212_SanJuan/ansible-lab/elastic-stack
kaizohost@kaizohost-VirtualBox...
kaizohost@kaizohost-VirtualBox...
kaizohost@kaizohost-VirtualBox...
kaizohost@kaizohost-VirtualBox...
TASK [kibana : Install Kibana] **********
****
changed: [192.168.56.106]
TASK [kibana : Configure Kibana] **********
****
changed: [192.168.56.106]
TASK [kibana : Enable and start Kibana] **********
****
changed: [192.168.56.106]
RUNNING HANDLER [kibana : restart kibana] **********
****
changed: [192.168.56.106]
PLAY [logstash] **********
****
TASK [Gathering Facts] **********
****
ok: [192.168.56.106]
TASK [logstash : Install Logstash] **********
****
changed: [192.168.56.106]
TASK [logstash : Configure pipeline] **********
****
changed: [192.168.56.106]
TASK [logstash : Enable and start Logstash] **********
****
changed: [192.168.56.106]
RUNNING HANDLER [logstash : restart logstash] **********
****

```
TASK [logstash : Install Logstash] *********************************************************************************************************************************
**************************************************
changed: [192.168.56.106]

TASK [logstash : Configure pipeline] *******************************************************************************************************************************
**************************************************
changed: [192.168.56.106]

TASK [logstash : Enable and start Logstash] ************************************************************************************************************************
**************************************************
changed: [192.168.56.106]

RUNNING HANDLER [logstash : restart logstash] **********************************************************************************************************************
**************************************************
changed: [192.168.56.106]

PLAY RECAP *********************************************************************************************************************************************************
**************************************************
192.168.56.106             : ok=17   changed=8    unreachable=0    failed=0    skipped=4    rescued=0    ignored=0
```

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab/elastic-stack$ ansible -i inventories/centos.ini all -m command -a "rpm -q elasticsearch
ibana logstash" -K -b
BECOME password:
192.168.56.106 | CHANGED | rc=0 >>
elasticsearch-8.19.23-1.x86_64
kibana-8.19.23-1.x86_64
logstash-8.19.23-1.x86_64
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab/elastic-stack$
```

2. Apply the concept of creating roles.

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab/elastic-stack$ tree roles
roles
├── common
│   ├── defaults
│   │   └── main.yml
│   ├── files
│   ├── handlers
│   │   └── main.yml
│   ├── meta
│   │   └── main.yml
│   ├── README.md
│   ├── tasks
│   │   └── main.yml
│   ├── templates
│   ├── tests
│   │   ├── inventory
│   │   └── test.yml
│   └── vars
│       └── main.yml
```


elasticsearch
defaults
main.yml
files
handlers
main.yml
meta
main.yml
README.md
tasks
main.yml
templates
elasticsearch.yml.j2
tests
inventory
test.yml
vars
main.yml
kibana
defaults
main.yml
files
handlers
main.yml
meta
main.yml
README.md
tasks
main.yml
templates
kibana.yml.j2
tests
inventory
test.yml
vars
main.yml



3. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)

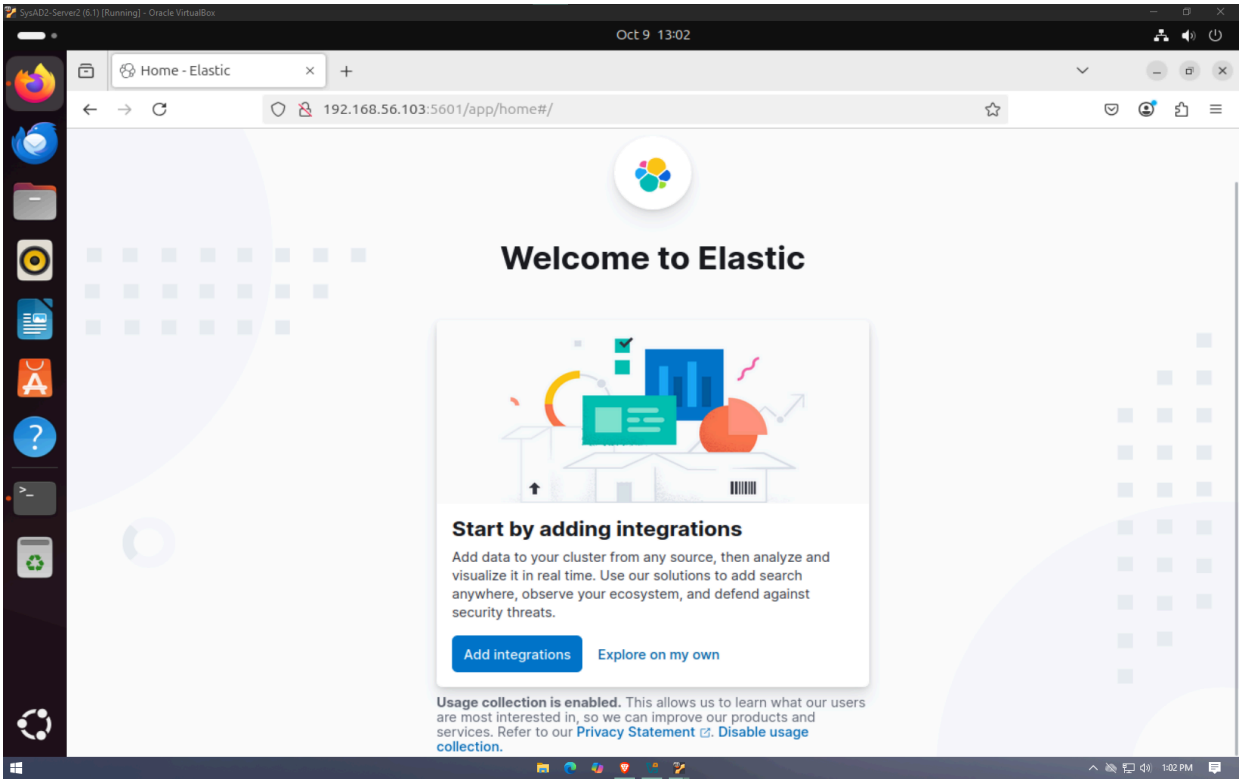
1. **Created the GitHub repo and set up Ansible. I made a new repo for the activity, cloned it to the control node, and installed Ansible.**
2. **Made the project folders. I created `elastic-stack` with `site.yml`, an `inventories/` folder, and a `roles/` folder.**
3. **Made one inventory per OS. I created `ubuntu.ini` and `centos.ini` with three groups (`elasticsearch`, `kibana`, `logstash`), so I could run one OS at a time. I tested the connection with `ansible -m ping`.**
4. **Created the roles. I used `ansible-galaxy init` to make `common`, `elasticsearch`, `kibana` and `logstash`.**
5. **Set up the `common` role. It adds the Elastic repo and GPG key. I used `ansible_os_family` so Ubuntu uses `apt` and CentOS uses `yum`.**
6. **Set up the service roles. Each role installs its package, copies a config from a template, and starts the service. A handler restarts the service when the config changes.**
7. **Wrote `site.yml`. It runs `common` on all hosts, then assigns each role to its host group.**
8. **Ran the playbook. I ran `ansible-playbook -i inventories/ubuntu.ini site.yml -K`, then did the same with `centos.ini`.**

9. **Fixed the errors. I fixed a wrong `-i` option, a stale apt cache, missing `path.data` and `path.logs` settings, and leftover TLS entries in the Elasticsearch keystore.**
10. **Verified the install. I used `curl` on ports 9200 and 5601 and `systemctl is-active` on both Ubuntu and CentOS.**

4. Show an output of the installed Elastic Stack for both Ubuntu and CentOS.
Ubuntu:

```
kaizoserver1@kaizoserver1-VirtualBox:~$ curl http://192.168.56.102:9200
{
  "name" : "192.168.56.102",
  "cluster_name" : "lab-cluster",
  "cluster_uuid" : "E-AuaAalQIC93hTBzb8eMA",
  "version" : {
    "number" : "8.19.22",
    "build_flavor" : "default",
    "build_type" : "deb",
    "build_hash" : "3b2a41103de35e0af4064d647974032fcc1bcde9",
    "build_date" : "2026-09-18T10:10:07.018982263Z",
    "build_snapshot" : false,
    "lucene_version" : "9.12.2",
    "minimum_wire_compatibility_version" : "7.17.0",
    "minimum_index_compatibility_version" : "7.0.0"
  },
  "tagline" : "You Know, for Search"
}
```

```
kaizoserver2@kaizoserver2-VirtualBox:~$ curl -I http://192.168.56.103:5601
HTTP/1.1 200 OK
x-content-type-options: nosniff
referrer-policy: strict-origin-when-cross-origin
permissions-policy: camera=(), display-capture=(), fullscreen=(self), geolocation=(), microphone=(), web-share=()
cross-origin-opener-policy: same-origin
content-security-policy: script-src 'report-sample' 'self'; worker-src 'report-sample' 'self' blob:; style-src 'report-sample' 'self' 'unsafe-inline'
content-security-policy-report-only: form-action 'report-sample' 'self'; default-src 'report-sample' 'none'; font-src 'report-sample' 'self'; img-src
'report-sample' 'self' data: tiles.maps.elastic.co; connect-src 'report-sample' 'self' telemetry.elastic.co telemetry-staging.elastic.co feeds.elastic
.co tiles.maps.elastic.co vector.maps.elastic.co
kbn-name: kaizoserver2-VirtualBox
content-type: text/html; charset=utf-8
cache-control: private, no-cache, no-store, must-revalidate
content-length: 103668
vary: accept-encoding
connection: close
Date: Fri, 09 Oct 2026 04:58:09 GMT
```




```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab/elastic-stack$ ansible -i inventories/ubuntu.ini logstash -m command -a "systemctl is-acti
ve logstash" -K
BECOME password:
192.168.56.103 | CHANGED | rc=0 >>
active
```

CentOS:

```
[server@localhost ~]$ curl http://192.168.56.106:9200
{
  "name" : "192.168.56.106",
  "cluster_name" : "lab-cluster",
  "cluster_uuid" : "iG7-KDxKQfadj2scSCihvw",
  "version" : {
    "number" : "8.19.23",
    "build_flavor" : "default",
    "build_type" : "rpm",
    "build_hash" : "6940e499f8a38841809b8e295a20c6421c95d9e4",
    "build_date" : "2026-09-30T23:11:55.493565832Z",
    "build_snapshot" : false,
    "lucene_version" : "9.12.2",
    "minimum_wire_compatibility_version" : "7.17.0",
    "minimum_index_compatibility_version" : "7.0.0"
  },
  "tagline" : "You Know, for Search"
}
```

```
[server@localhost ~]$ curl -I http://192.168.56.106:5601
HTTP/1.1 302 Found
location: /spaces/enter
x-content-type-options: nosniff
referrer-policy: strict-origin-when-cross-origin
permissions-policy: camera=(), display-capture=(), fullscreen=(self), geolocation=(), microphone=(), web-share=()
cross-origin-opener-policy: same-origin
content-security-policy: script-src 'report-sample' 'self'; worker-src 'report-sample' 'self' blob:; style-src 'report-sample' 'self' 'unsafe-inline'
content-security-policy-report-only: form-action 'report-sample' 'self'; default-src 'report-sample' 'none'; font-src 'report-sample' 'self'; img-src 'rep
sample' 'self' data: tiles.maps.elastic.co; connect-src 'report-sample' 'self' telemetry.elastic.co telemetry-staging.elastic.co feeds.elastic.co tiles.ma
lastic.co vector.maps.elastic.co
kbn-name: localhost.localdomain
kbn-license-sig: a742dbe3793d9239a891ecc66ec939598075217fc4d0a3497602f48eb56603da
cache-control: private, no-cache, no-store, must-revalidate
content-length: 0
Date: Fri, 09 Oct 2026 05:32:20 GMT
Connection: keep-alive
Keep-Alive: timeout=120
```

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab/elastic-stack$ ansible -i inventories/centos.ini all -m command -a "systemctl is-active el
asticsearch kibana logstash" -K
BECOME password:
192.168.56.106 | CHANGED | rc=0 >>
active
active
active
```

5. Make sure to create a new repository in GitHub for this activity

https://github.com/ivnwynn/CPE212_SanJuan/tree/main/ansible-lab/elastic-stack

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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